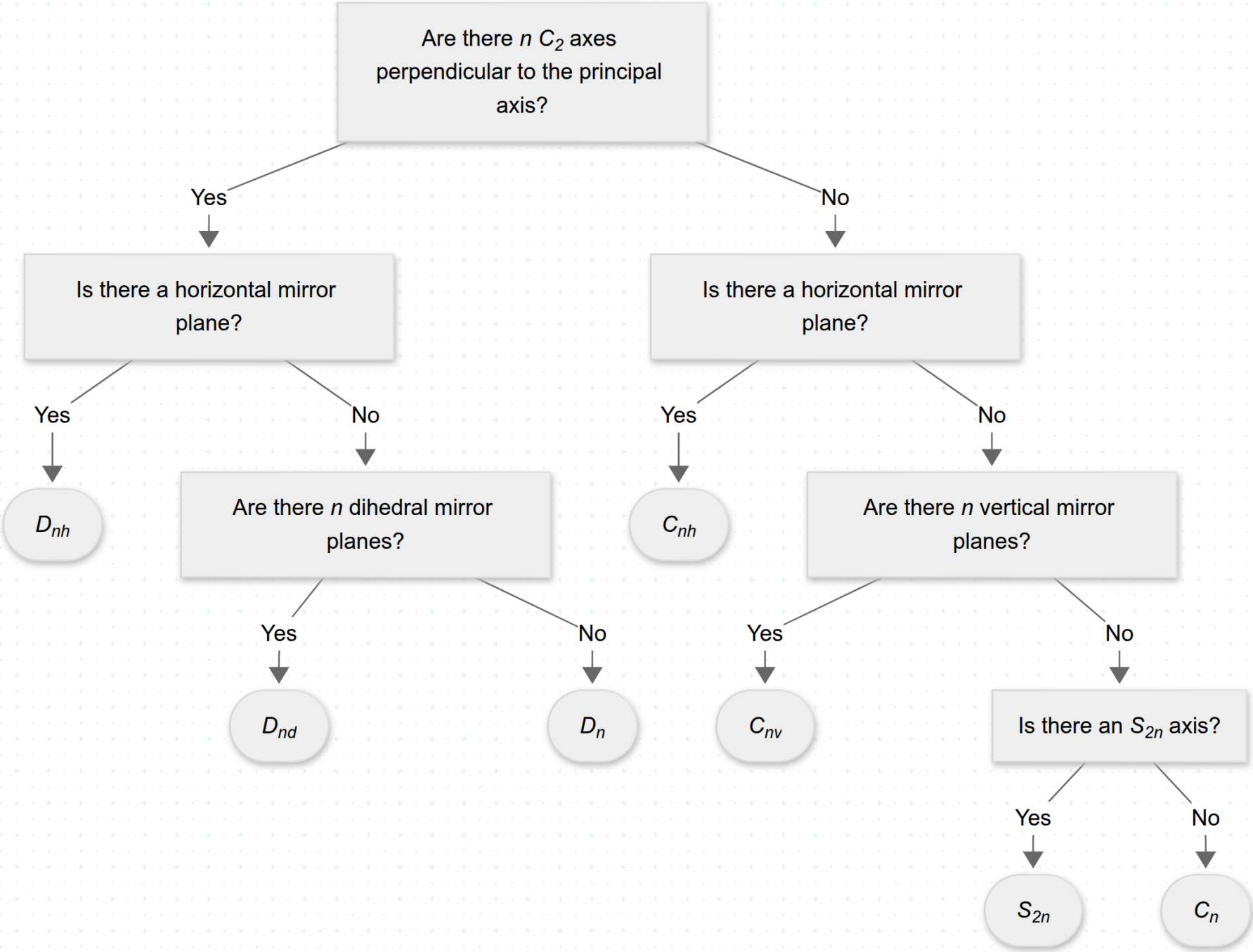
Are there n C_2 axes perpendicular to the principal axis?
Yes
No
Is there a horizontal mirror plane?
Is there a horizontal mirror plane?
Yes
No
Yes
No
D_{nh}
Are there n dihedral mirror planes?
C_{nh}
Are there n vertical mirror planes?
Yes
No
Yes
No
D_{nd}
D_n
C_{nv}
Is there an S_{2n} axis?
Yes
No
S_{2n}
C_n

C_{2v}	E	C_2	σ_{v1}	σ_{v2}
A_1	1	1	1	1
A_2	1	1	-1	-1
B_1	1	-1	1	-1
B_2	1	-1	-1	1

C_{3v}	E	$2C_3$	$3\sigma_v$
A_1	1	1	1
A_2	1	1	-1
E	2	-1	0

C_{5v}	E	$2C_5$	$2C_5^2$	$5\sigma_v$
A_1	1	1	1	1
A_2	1	1	1	-1
E_1	2	$-\frac{1}{2}+\frac{\sqrt{5}}{2}$	$-\frac{1}{2}-\frac{\sqrt{5}}{2}$	0
E_2	2	$-\frac{1}{2}-\frac{\sqrt{5}}{2}$	$-\frac{1}{2}+\frac{\sqrt{5}}{2}$	0

D_5	E	$2C_5$	$2C_5^2$	$5C_2$
A_1	1	1	1	1
A_2	1	1	1	-1
E_1	2	$-\frac{1}{2}+\frac{\sqrt{5}}{2}$	$-\frac{1}{2}-\frac{\sqrt{5}}{2}$	0
E_2	2	$-\frac{1}{2}-\frac{\sqrt{5}}{2}$	$-\frac{1}{2}+\frac{\sqrt{5}}{2}$	0

D_{5d}	E	$2C_5$	$2C_5^2$	$5C_2$	i	$2S_{10}^3$	$2S_{10}$	$5\sigma_d$
A_{1g}	1	1	1	1	1	1	1	1
A_{2g}	1	1	1	-1	1	1	1	-1
E_{1g}	2	$-\frac{1}{2}+\frac{\sqrt{5}}{2}$	$-\frac{1}{2}-\frac{\sqrt{5}}{2}$	0	2	$-\frac{1}{2}+\frac{\sqrt{5}}{2}$	$-\frac{1}{2}-\frac{\sqrt{5}}{2}$	0
E_{2g}	2	$-\frac{1}{2}-\frac{\sqrt{5}}{2}$	$-\frac{1}{2}+\frac{\sqrt{5}}{2}$	0	2	$-\frac{1}{2}-\frac{\sqrt{5}}{2}$	$-\frac{1}{2}+\frac{\sqrt{5}}{2}$	0
A_{1u}	1	1	1	1	-1	-1	-1	-1
A_{2u}	1	1	1	-1	-1	-1	-1	1
E_{1u}	2	$-\frac{1}{2}+\frac{\sqrt{5}}{2}$	$-\frac{1}{2}-\frac{\sqrt{5}}{2}$	0	-2	$\frac{1}{2}-\frac{\sqrt{5}}{2}$	$\frac{1}{2}+\frac{\sqrt{5}}{2}$	0
E_{2u}	2	$-\frac{1}{2}-\frac{\sqrt{5}}{2}$	$-\frac{1}{2}+\frac{\sqrt{5}}{2}$	0	-2	$\frac{1}{2}+\frac{\sqrt{5}}{2}$	$\frac{1}{2}-\frac{\sqrt{5}}{2}$	0

C_s	E	σ_h
A'	1	1
A''	1	-1

C_5	E	C_5	C_5^2	C_5^3	C_5^4
A	1	1	1	1	1
E_1	1	ε	ε^2	$\overline{\varepsilon^2}$	$\overline{\varepsilon}$
	1	$\overline{\varepsilon}$	$\overline{\varepsilon^2}$	ε^2	ε
E_2	1	ε^2	$\overline{\varepsilon}$	ε	$\overline{\varepsilon^2}$
	1	$\overline{\varepsilon^2}$	ε	$\overline{\varepsilon}$	ε^2

D_{5h}	E	$2C_5$	$2C_5^2$	$5C_2$	σ_h	$2S_5$	$2S_5^3$	$5\sigma_v$
A_1'	1	1	1	1	1	1	1	1
A_2'	1	1	1	-1	1	1	1	-1
E_1'	2	$-\frac{1}{2}+\frac{\sqrt{5}}{2}$	$-\frac{1}{2}-\frac{\sqrt{5}}{2}$	0	2	$-\frac{1}{2}+\frac{\sqrt{5}}{2}$	$-\frac{1}{2}-\frac{\sqrt{5}}{2}$	0
E_2'	2	$-\frac{1}{2}-\frac{\sqrt{5}}{2}$	$-\frac{1}{2}+\frac{\sqrt{5}}{2}$	0	2	$-\frac{1}{2}-\frac{\sqrt{5}}{2}$	$-\frac{1}{2}+\frac{\sqrt{5}}{2}$	0
A_1''	1	1	1	1	-1	-1	-1	-1
A_2''	1	1	1	-1	-1	-1	-1	1
E_1''	2	$-\frac{1}{2}+\frac{\sqrt{5}}{2}$	$-\frac{1}{2}-\frac{\sqrt{5}}{2}$	0	-2	$\frac{1}{2}-\frac{\sqrt{5}}{2}$	$\frac{1}{2}+\frac{\sqrt{5}}{2}$	0
E_2''	2	$-\frac{1}{2}-\frac{\sqrt{5}}{2}$	$-\frac{1}{2}+\frac{\sqrt{5}}{2}$	0	-2	$\frac{1}{2}+\frac{\sqrt{5}}{2}$	$\frac{1}{2}-\frac{\sqrt{5}}{2}$	0

C_3	E	C_3	C_3^2
A	1	1	1
E_1	1	ε	$\overline{\varepsilon}$
	1	$\overline{\varepsilon}$	ε